AlignΞd

★ 4.9/5 (173)

£98.00

About

Lightweight, supportive knee alignment system designed for beginner gym users and lower-body training. ***AlignΞd*** uses IMU motion sensing and targeted haptic feedback to help users recognise poor knee alignment, improve squat awareness, and train with more confidence.

Learn more

Add to bag - £98.00

Product Type

Training Support

A wearable support system for gym beginners who want clearer movement guidance and greater confidence during lower-body training.

Ideal For

Training Scenarios

Beginner gym users, lower-body training, squat practice, and users who want clearer movement guidance during exercise.

What's included +

Full specifications +

Survey and Research

Interviewed 20 participants and made further research about the squat

Participant Voices

We interviewed gym beginners with varied ages and training experience, alongside a gym trainer.

The discussion **focused on** confidence and challenges around squat-based training.

“I often feel unsure about my form and worry that others may judge me.”

“I would feel more confident if I had simple guidance during training.”

“Poor balance, incorrect knee positioning and weak control of the centre of mass can lead to injury.”

“Beginners often struggle most with weighted squats and deadlifts.”

88% experience

gym anxiety

A Gymshark study of 1,000 women found that 88% reported experiencing some form of anxiety in the gym. This supports a design direction focused on confidence-building, helping beginner users feel more comfortable when entering the gym

12.6% injury prevalence

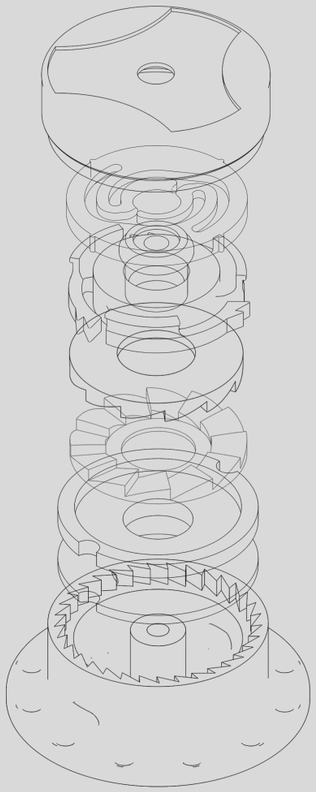
in strength training

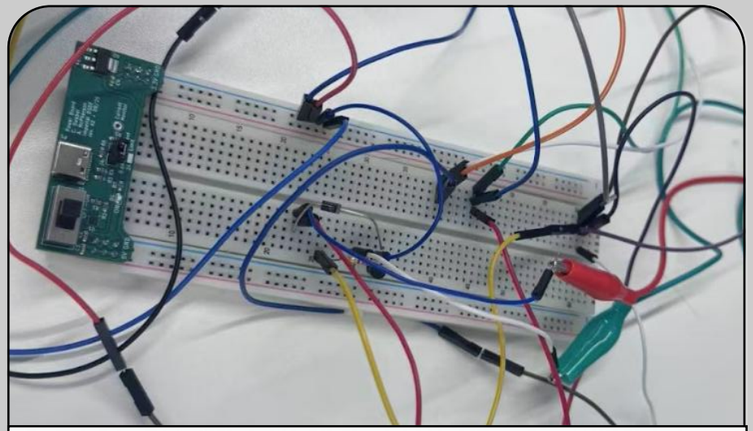
systematic review found that traditional strength training had a mean injury prevalence of 12.6%.This suggests the product should support beginners before they progress to heavier training, helping them build safer movement habits early.

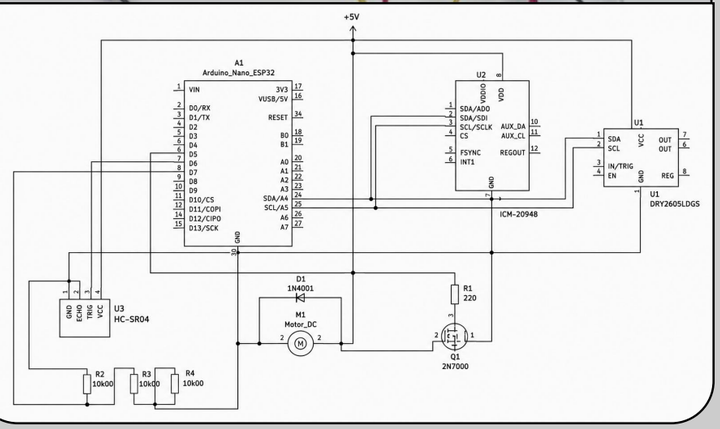
27% cannot complete

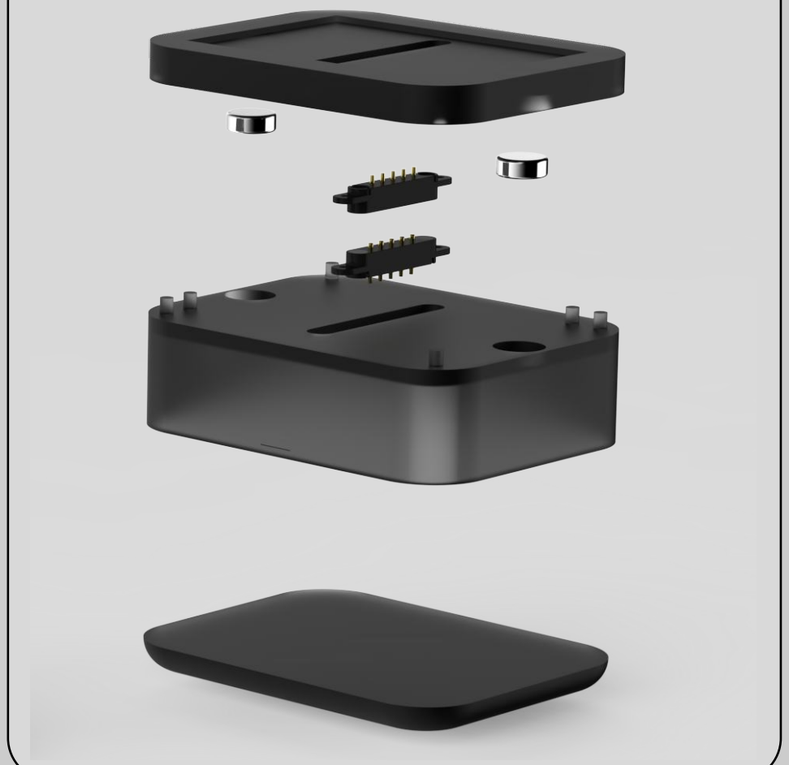
a single squat

A PureGym survey found that over a quarter of the population were unable to complete a single squat. This supports a product direction focused on beginner lower-body training, helping users build correct movement habits from the start.









Mechanical System

The BOA tightening system provides quick and precise fit adjustment for the knee brace. Users can press and rotate the dial to gradually tighten the support.The internal gear mechanism converts this rotation into controlled tension, keeping the brace secure and stable

Electrical System

The electrical system combines motion sensing, processing and haptic feedback into one compact module. TheIMU captures knee movement data, while the ESP32-S2 processes the signal and triggers the DRV2605L haptic driver.

Attachment System

Pogo pins and two opposite-pole magnets allow the pod to attach in only one correct orientation. The pogo pins connect the embedded vibration motors to the removable pod, where the battery and controller power and trigger the haptic feedback system

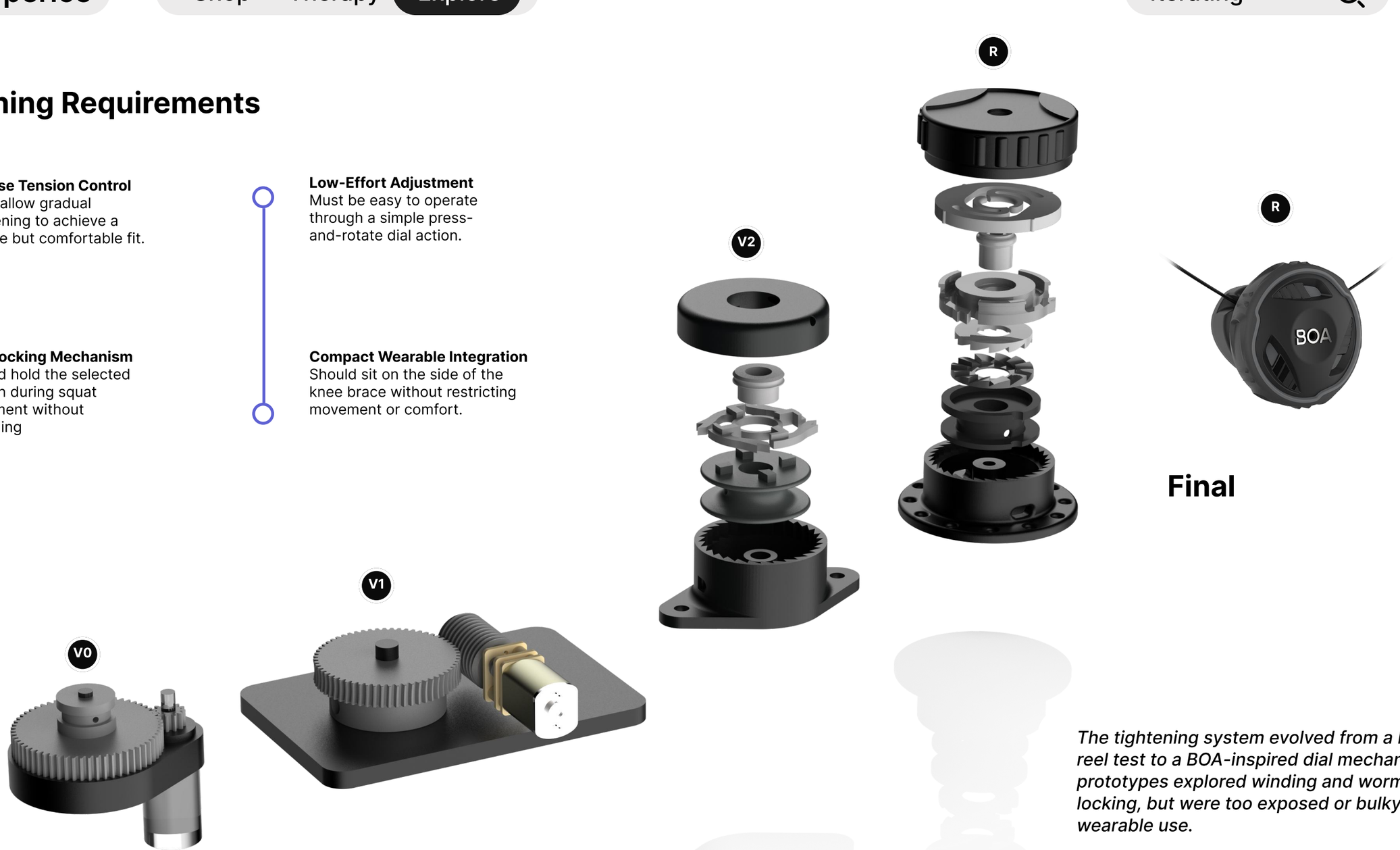
Tightening Requirements

Precise Tension Control
Must allow gradual tightening to achieve a secure but comfortable fit.

Low-Effort Adjustment
Must be easy to operate through a simple press-and-rotate dial action.

Self-Locking Mechanism
Should hold the selected tension during squat movement without loosening

Compact Wearable Integration
Should sit on the side of the knee brace without restricting movement or comfort.



Final

The tightening system evolved from a basic wire reel test to a BOA-inspired dial mechanism. Early prototypes explored winding and worm gear locking, but were too exposed or bulky for wearable use.

V0 — **Wire reel**
Basic servo winding test.

V1 — **Worm gear**
Passive locking exploration.

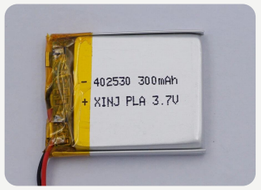
V2 — **Enclosed dial**
Compact gear housing.

R — **BOA research**
Industry fit-system study.

V3 — **BOA recreation**
Brace-side tightening logic.

Hardware Components

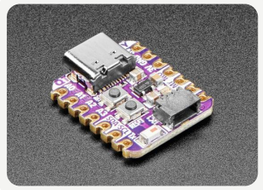
Core electronics and haptic parts selected for the Hyperice prototype, with compact voltage/current estimates.



Battery

3.7V nominal / 300mAh / ~1.11Wh

Rechargeable lithium cell, compact and widely used within small electronic products

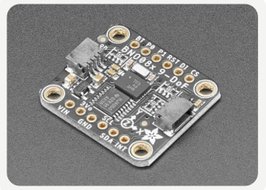


Adafruit QT Py ESP32-S2

3.3V logic / sleep ~70uA / active 12-32mA / Wi-Fi peak up to 310mA

Compact Wi-Fi microcontroller board

board/chip-state dependent estimate

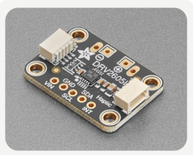


BNO055 IMU

VDD 2.4-3.6V / VDDIO 1.7-3.6V / ~12.3mA

9-axis absolute orientation sensor

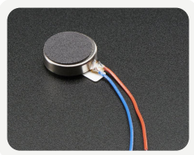
visual reference; BNO055 electrical values



DRV2605L haptic motor controller

VDD 2-5.2V / 0.5-0.65mA idle / 2.3-3.5mA no-load haptic

I2C haptic driver for ERM and LRA actuators

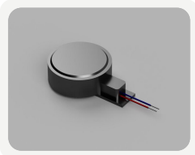


1.8V coin vibration motor

1.8V ERM / ~36-40mA running estimate

Standard coin-style eccentric rotating mass actuator

proxy estimate; load-test required



Coin-style LRA motor

LRA actuator / load-dependent current / ~80mA proxy estimate

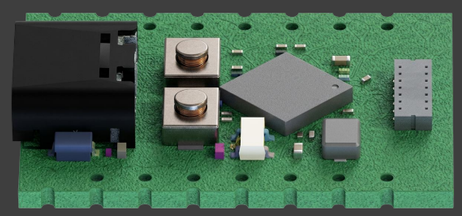
Linear resonant actuator for controlled haptic feedback

provisional until load testing

Pogo pin

Passive connector / no active current draw

Spring contact for charging, testing, or board-to-fixture connection. Snaps magnetically



Custom PCB integration

The electronics are consolidated around a compact custom PCB. The controller, IMU, haptic driver, connectors, battery input, and motor outputs are arranged as one integrated assembly.

Power and Lifetime

A 3.7V / 300mAh battery provides 1.11Wh nominal energy. Estimates combine QT Py active draw, BNO055 normal mode, and DRV2605L idle current; ERM and LRA haptic loads are modeled as separate active scenarios.

Sources: TI DRV2605L / Bosch BNO055 / Espressif ESP32-S2 / Adafruit QT Py + motor supplier pages

Power Calculation

Battery energy: 3.7V x 0.300Ah = 1.11Wh

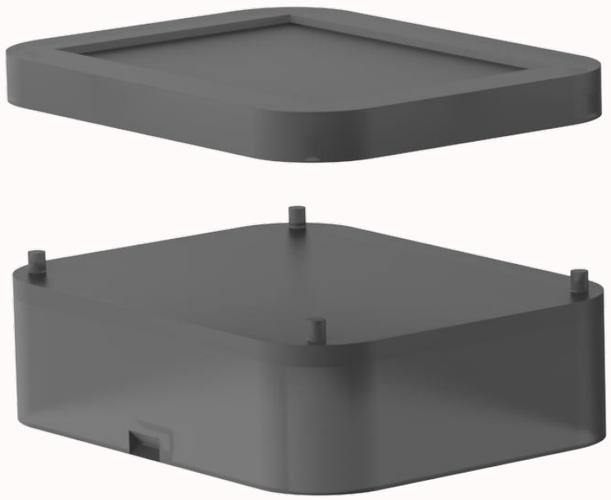
Electronics baseline: ~33mA / ~0.12W / ~9.1h

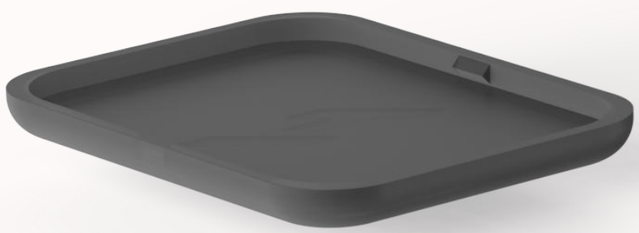
ERM haptic active: ~73mA / ~0.27W / ~4.1h

LRA haptic active: ~113mA / ~0.42W / ~2.7h

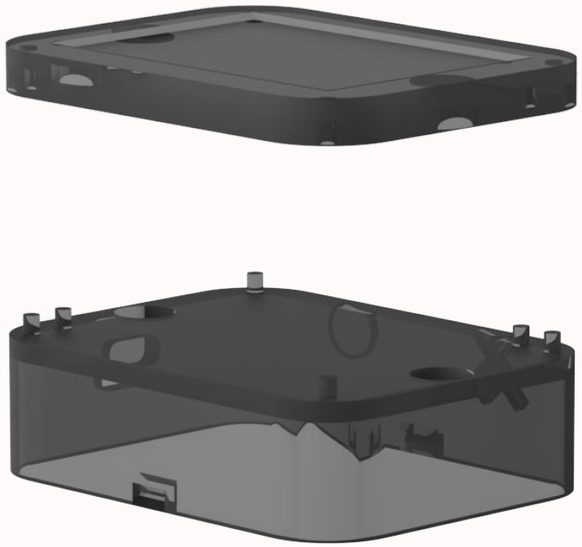
Baseline = QT Py active + BNO055 normal + DRV idle. ERM/LRA not simultaneous; motor current and final runtime remain provisional until load testing.

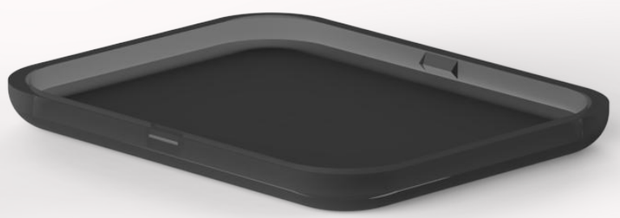
V1 - simple box to house electronics





V2 - asymmetrical alignment pegs and body refinement





V3 - pogo pin holes and lid modification

Tightening Mechanism

Each part can be injection moulded. ABS, Nylon and Acetal used for body, moving components and the spring.

Chamfering

External and internal chamfering strengthen the build whilst reducing the number of components needed, saving time and reducing complexity, whilst maintaining aesthetics.

Magnetic connections

The base and body can be made from one type of rigid plastic- use of magnets and pegs in place of snap fits eliminate the need for flexible plastics, which would also lose their strength more easily with repeated use.

Wiring connector

Components housed within the brace all meet at a central pogo pin connector

Hyperice Logo

Minimalist, 3D inset coloured logo pops enough to draw attention to the brand, whilst keeping to its core minimalist style.

Inter-changeable, colour represents subdivisions such as their red “Venom” lineup.

PCB

Can be produced repetitively and in bulk, reducing error, unit costs and production time. Also allows components to be housed in a smaller pod, saving resources and money.

Print Optimisation

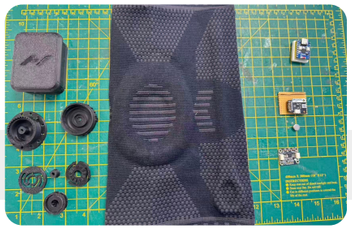
Oriented to minimise support material and warping during printing. Tolerance of 0.2mm for the snap fits, and are identical but laid out asymmetrically.

ASSEMBLY Process

Photo-led assembly sequence with timing and cost estimates

Time + Cost

01



Knee Brace Body Preparation

Mark BOA, pod interface and wire routing.

2 min / $0.05

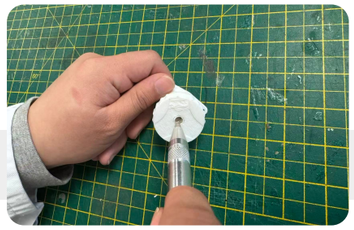
02

Pod Positioning Decision

Avoid bending area and allow easy removal.

2 min / $0.00

03

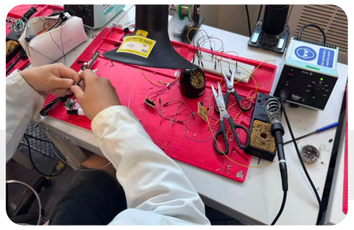


BOA Tightening System Installation

Install adjustable tightening system.

3min / $0.30

04

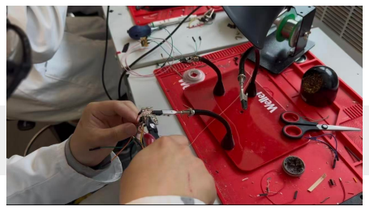


Embedding and Soldering

Embed motor and solder insulated wires.

6min / $0.45

05

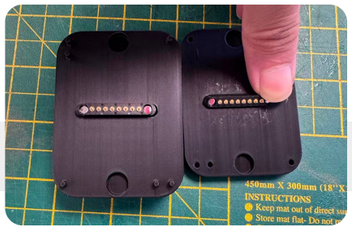


Internal Wire Routing

Route motor wires through textile layers.

3min / $0.15

06



Pogo Pin Interface Installation

Install detachable contact interface.

2min / $0.60

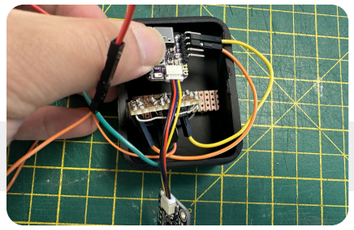
07

Magnetic Alignment Assembly

Add magnets for alignment and pod security.

1min / $0.40

08



3D Printed Electronics Box Assembly

Assemble IMU, controller, battery and circuit.

4 min / $0.75

09

Pogo Pin to Controller Connection

Connect pogo terminals to controller.

3 min / $0.10

10

Pod Attachment and Contact Test

Check magnetic alignment, contact and stability.

2 min / $0.00

11

Haptic Feedback Function Test

Confirm sensing, response and vibration feedback.

3min / $0.00

12

Final Fit and Safety Check

Check comfort, wire safety and product function.

3min / $0.00

Total Time for Assembly: 34 Minutes

Total Cost of Assembly: $2.80

estimated timing + consumables

Train Hard

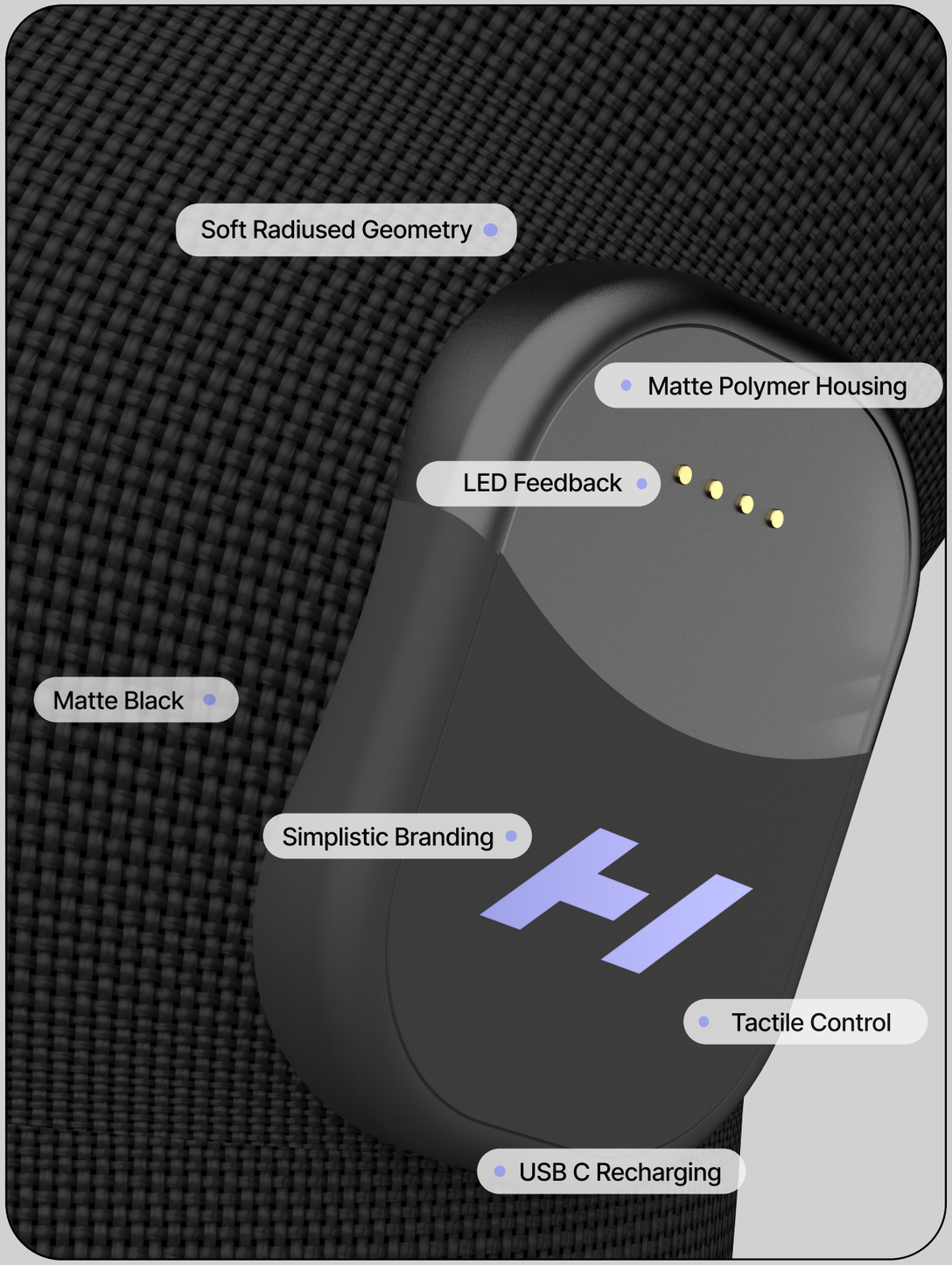
STAY ALIGNED

Make Your Style ....

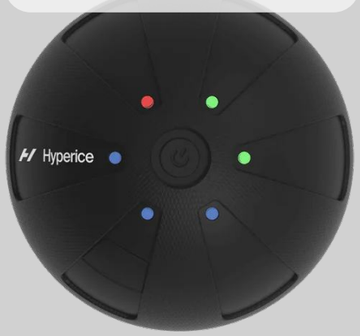
Technical Knit

Matte Technical Textile

Rounded Module Geometry



Colour LED Feedback



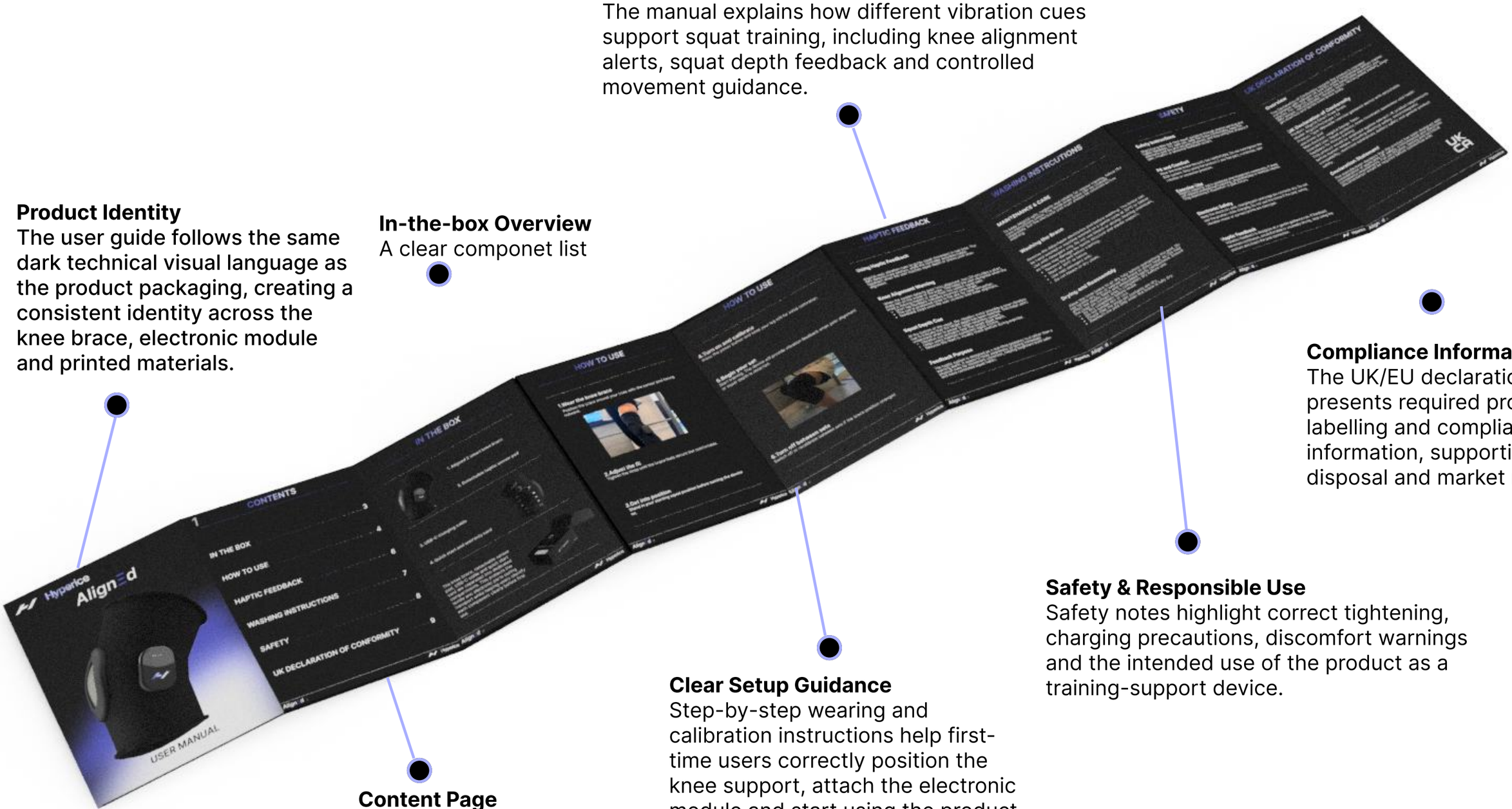
Tactile Button Interface

Compact Soft-Edge Form

Haptic Feedback Explanation
The manual explains how different vibration cues support squat training, including knee alignment alerts, squat depth feedback and controlled movement guidance.

Product Identity
The user guide follows the same dark technical visual language as the product packaging, creating a consistent identity across the knee brace, electronic module and printed materials.

In-the-box Overview
A clear componet list



Content Page
talk about the directory

Clear Setup Guidance
Step-by-step wearing and calibration instructions help first-time users correctly position the knee support, attach the electronic module and start using the product with minimal confusion.

Safety & Responsible Use
Safety notes highlight correct tightening, charging precautions, discomfort warnings and the intended use of the product as a training-support device.

Compliance Information
The UK/EU declaration page presents required product labelling and compliance information, supporting safe use, disposal and market readiness.

Premium Outer Box

Matte black paperboard, minimal Hyperice logo placement and clean white typography create a premium sports-tech appearance. The restrained colour palette keeps the packaging consistent with Hyperice's performance-focused brand identity.

Consistent Product Identity

The AlignEd name, product render, feature icons and typography are applied consistently across the outer box, product label and user guide.

Guided First-Use Layout

The knee support, detachable electronics and charging accessories are separated into clear zones, helping first-time users understand each component during unboxing.

Protective Internal Fitment

Folded cardboard inserts hold the brace and electronic modules securely during transport, reducing movement and impact without using plastic trays.

Recyclable Material System

Paper-based packaging and recyclable inserts reduce material complexity while maintaining protection, lightness and a premium finish.

1. Detachable Electronics Pod
- Houses IMU sensor, microcontroller and battery
- Reads knee posture during squat movement
- Removable before washing

2. Pogo Pin + Magnetic Interface
- Creates detachable electrical connection
- Powers embedded vibration motor only when pod is attached
- Magnets ensure correct orientation and stable attachment

3. BOA Tightening System
- Allows quick fit adjustment
- Improves brace stability during exercise
- Helps maintain accurate sensor positioning

4. Embedded Haptic Feedback Motor
- Positioned close to the skin inside the brace
- Provides clear vibration cues
- Alerts user to poor knee alignment or squat movement errors

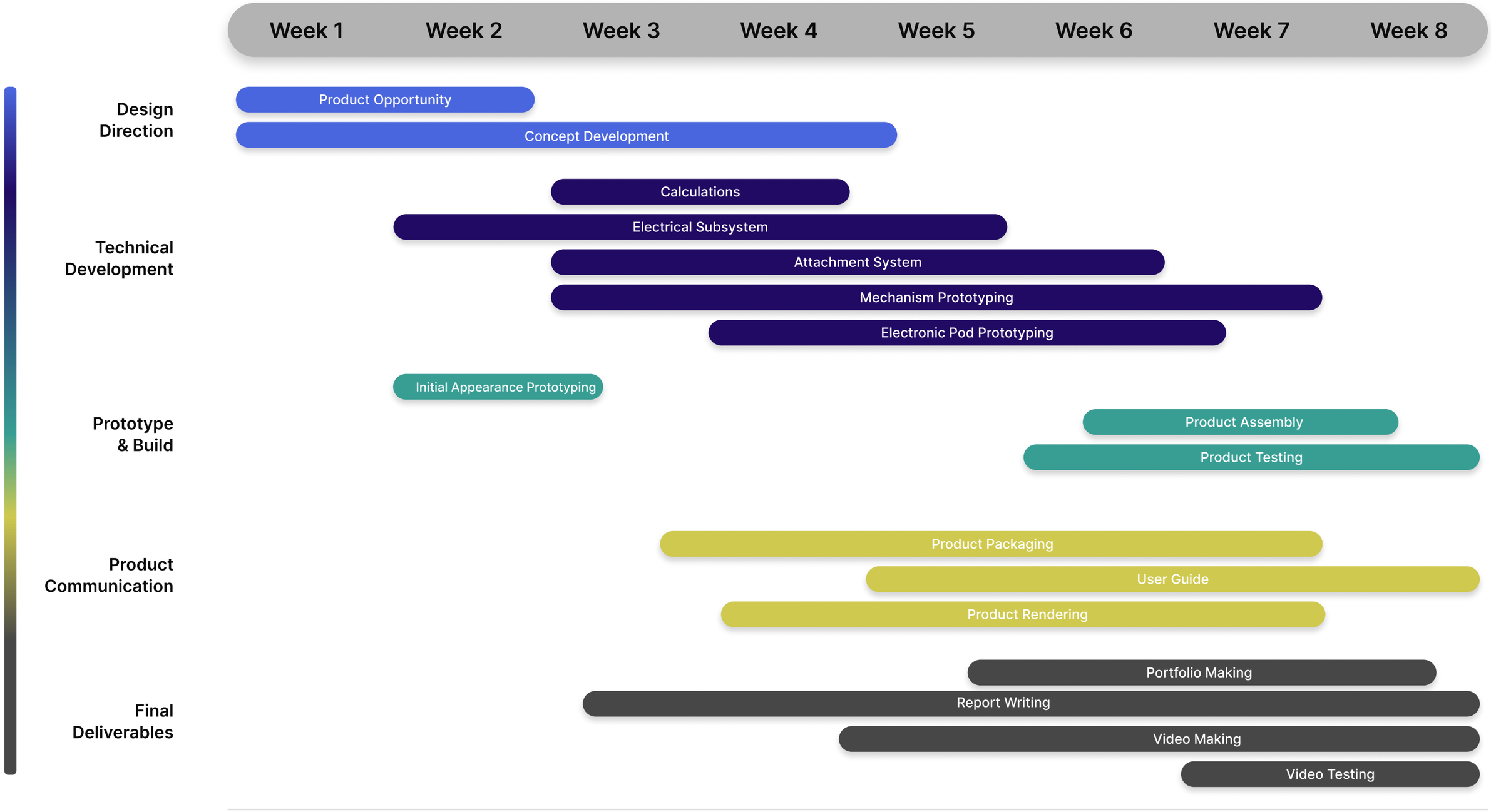
Aligned

The confidence to load the bar

Hyperice

Project Timeline

A Gantt chart was used to monitor progress across the eight-week Phase 02 development schedule.



Colour key — Design Direction — Technical Development — Prototype & Build — Product Communication — Final Deliverables